

Patent Intelligence Project

Model Selection for Semantic Patent Retrieval

1 Suitable Models

1.1 TF-IDF / BM25 (Lexical Baseline)

Advantages:

- Trivial to implement with `scikit-learn`; no additional heavy dependencies beyond what is already installed.
- Extremely fast to build and query, even over the full 25k-document sample.
- Fully interpretable; useful as a sanity-check baseline to quantify how much semantic models improve over pure keyword overlap.

Disadvantages:

- Purely lexical — cannot match paraphrased inventions, which is precisely the failure mode the project was created to solve.
- No notion of synonymy, hypernymy, or contextual meaning.

1.2 Static Word Embeddings (Word2Vec / GloVe, Averaged)

Advantages:

- Lightweight, fast inference, easy to train custom vectors on patent-domain text if desired.
- Some improvement over pure lexical matching (captures loose synonymy).

Disadvantages:

- Averaging word vectors discards word order and syntactic structure, which matters in claim language.
- Static vectors cannot disambiguate polysemous technical terms.
- Generally outperformed by transformer-based sentence embeddings on retrieval benchmarks.

1.3 Doc2Vec

Advantages:

- Produces document-level vectors directly, without a separate pooling step.
- Unsupervised; trains directly on the HUPD corpus without external checkpoints.

Disadvantages:

- Embedding quality is generally weaker than transformer-based methods on modern benchmarks.
- Sensitive to hyperparameters and corpus size; 16k training documents is on the small side for training Doc2Vec from scratch.
- Little patent-specific tooling or precedent to build on.

1.4 Generic BERT (Pooled [CLS] or Mean-Pooled)

Advantages:

- Strong contextual understanding of language; widely supported via Hugging Face **transformers**, which is already in the project’s dependency stack.

Disadvantages:

- Raw BERT embeddings are known to perform poorly on semantic similarity tasks without a fine-tuning step specifically for that purpose (anisotropic embedding space).
- 512-token limit is short relative to full patent claims/description fields, forcing aggressive truncation or chunking.

1.5 Sentence-BERT Family (e.g. all-mpnet-base-v2, all-MiniLM-L6-v2)

Advantages:

- Purpose-built (via siamese/triplet training) to produce directly comparable sentence/document embeddings for cosine similarity — exactly the operation the retrieval pipeline needs.
- Fast inference; lightweight variants (MiniLM) trade a little accuracy for large speed gains.
- Mature integration with vector search libraries (FAISS, etc.) and extensive documentation.

Disadvantages:

- Trained predominantly on general-domain text, not patent language or claim structure.
- Still limited to roughly 384–512 tokens per input, so long **description** fields require chunking and aggregation.

1.6 Domain-Specific Patent Embedding Models (PatentSBERTa, Patent-BERT)

Advantages:

- Fine-tuned specifically on patent text and patent classification codes (CPC/IPC), so they already understand patent-specific vocabulary and claim phrasing.
- Directly aligned with the project’s task: published evaluations report improved patent-to-patent similarity performance over generic SBERT models.

Disadvantages:

- Smaller community and maintenance footprint than mainstream SBERT models; checkpoint availability and long-term support are less certain.
- Fewer size/speed variants to choose from.

1.7 SPECTER / SPECTER2

Advantages:

- Designed from the ground up for document-level (not sentence-level) embeddings, using a citation-informed training signal — conceptually close to the “related prior art” relationship this project wants to model.
- Actively maintained by AllenAI, well documented, freely available on Hugging Face.
- Strong published performance on scientific/technical document retrieval tasks.

Disadvantages:

- Trained primarily on scientific papers rather than patents; some domain gap remains and may benefit from later fine-tuning on HUPD.
- Still BERT-based under the hood, so it shares the moderate context-length ceiling.

1.8 Long-Context Transformers (Longformer, BigBird, long-context E5/GTE variants)

Advantages:

- Can encode thousands of tokens directly, avoiding chunking for long **description** fields.

Disadvantages:

- Heavier compute and memory footprint than SBERT-style models, a real concern without a dedicated GPU.
- Less mature as off-the-shelf sentence/document embedding models compared to SBERT-style checkpoints.
- Likely unnecessary if the shorter, information-dense fields (abstract, claims) already carry most of the retrieval signal.

1.9 Modern General-Purpose Embedding Models (BGE, E5, GTE)

Advantages:

- State-of-the-art general semantic retrieval performance (top of public embedding benchmarks), actively maintained.
- Multiple sizes available, allowing a speed/accuracy trade-off suited to local hardware.
- Strong out-of-the-box performance without any fine-tuning.

Disadvantages:

- General-domain training, not patent-specific.
- Context windows vary by variant (512 up to a few thousand tokens) and must be checked per model.

1.10 Commercial Embedding APIs (OpenAI, Voyage AI, Cohere)

Advantages:

- Very strong general performance with no local GPU requirement.
- Simple API integration; no model management.

Disadvantages:

- Recurring cost that scales with corpus size (25k+ records already, likely to grow).
- Sends patent text to a third party — a real confidentiality/IP consideration for a patent-focused project.
- Reproducibility and offline development are harder to guarantee.

2 Comparison Summary

Model	Key Strength	Key Weakness for This Project
TF-IDF / BM25	Fast, simple baseline	No semantic matching
Word2Vec / GloVe	Lightweight, cheap	Loses order/context
Doc2Vec	Direct document vectors	Weaker quality, small training corpus
Generic BERT	Strong contextual model	Not tuned for similarity; short context
Sentence-BERT	Built for similarity search, fast	General-domain, not patent-tuned
PatentSBERTa / PatentBERT	Patent-domain tuned	Smaller community, availability risk
SPECTER / SPECTER2	Document-level, retrieval-oriented, maintained	Sci-paper domain, needs adaptation
Long-context models	Handles full-length text	Heavy compute, less mature tooling

Model	Key Strength	Key Weakness for This Project
BGE / E5 / GTE	Best general-purpose accuracy	Not patent-specific
Commercial APIs	Strong, zero local compute	Cost, sends IP text externally

3 Best Model to Use

SPECTER2 is recommended as the primary embedding model for this stage of the project, with **Sentence-BERT** (`all-mpnet-base-v2`) kept as a fast, general-purpose baseline to quantify improvement against, and **PatentSBERTa** flagged as a strong follow-up candidate once its checkpoint availability and licensing are confirmed.

Rationale:

- SPECTER2 embeds at the *document* level rather than the sentence level, matching how the retrieval task is actually framed (comparing whole inventions, not isolated sentences).
- Its citation-informed training objective is conceptually the closest available analogue to a “related prior art” relationship, among freely available, well-maintained checkpoints.
- It runs locally through the already-installed Hugging Face stack, with no API cost and no external data transfer — appropriate given the confidentiality of patent text.
- It is actively maintained with clear documentation, reducing the environment-compatibility risk already experienced once in this project (Stage 1, `datasets` version conflict).

This is a starting-point decision for the current development phase, not a final judgment. As with the HUPD dataset-selection decision in Stage 2, later fine-tuning of SPECTER2 (or a switch to PatentSBERTa) directly on HUPD claims/abstracts remains an open option once a baseline retrieval pipeline is working end to end.